Algorithm Section

Step-by-step procedure:

1. Collect user input (room type, date, time)
2. Process the input using chatbot interface
3. Validate availability of rooms
4. If available → confirm booking
5. If not → suggest alternative slots
6. Store booking details in database
7. Display confirmation message

Logic or method used:

- Rule-based + NLP-based chatbot logic
- Uses keyword matching and intent recognition

Model or technique applied:

- Natural Language Processing (NLP)
- IBM Watson Assistant / Dialogflow chatbot model

Process flow of the system:

User Input → Chatbot → Intent Detection → Database Check → Response Generation → Booking Confirmation

Dataset Section

Source of data:

- Sample dataset created manually / college room data

Number of records/samples:

- Around 100–500 records

Features or attributes (columns):

- Room ID
- Room Type
- Availability Status
- Date
- Time Slot
- Booking ID

Training and testing data split:

- 80% Training Data
- 20% Testing Data

Data format or type:

- CSV / JSON format